

**NMAM INSTITUTE OF TECHNOLOGY, NITTE**

Off-Campus Centre of Nitte (Deemed to be University)

Third Semester B.Tech (CBCS) Degree Examinations

December 2025

AD1102-1 – FUNDAMENTALS OF DATA SCIENCE

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **one full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. **What is the primary goal of Data Science?**
A) To store data
B) To create databases
C) To extract knowledge and insights from data
D) To write programs
2. **Which of the following best describes "big data"?**
A) Data stored in Excel files
B) Large and complex datasets that cannot be handled by traditional tools
C) Data that is generated by a small group of users
D) Data that is always structured and well-organized
3. **What does a data scientist primarily do?**
A) Writes web applications
B) Cleans, analyzes, and interprets data to help solve problems
C) Manages hardware infrastructure
D) Designs user interfaces for applications
4. **Which of the following is a subfield of Data Science that focuses on algorithms that learn from data?**
A) Data Visualization
B) Machine Learning
C) Cloud Computing
D) Network Security
5. **In which phase of the Data Science process is "cleaning data" most important?**
A) Data Collection
B) Model Deployment
C) Data Exploration
D) Data Preprocessing
6. **What is the primary purpose of Statistical Inference?**
A) To collect data from the population
B) To conduct experiments
C) To make conclusions about a population based on a sample
D) To visualize data trends
7. **Which of the following is an example of a point estimate?**
A) A 95% confidence interval
B) A hypothesis test result
C) A p-value
D) The sample mean
8. **Which of the following is NOT a measure of central tendency?**
A) Mean
B) Mode
C) Median
D) Variance
9. **Which of the following is used to measure the spread or dispersion of data?**
A) Mean
B) Standard deviation
C) Mode
D) Percentile
10. **What is the arithmetic mean of the following data: 10, 15, 20, 25, 30?**
A) 15
B) 22
C) 18
D) 20
11. **If a dataset has a high standard deviation, it indicates that:**
A) Data points are close to the mean
B) The dataset is skewed
C) Data points are spread out from the mean
D) There are outliers in the dataset

12. What does a positive skewness value indicate about a dataset?
 A) The data is symmetrical
 B) The data has a long right tail
 C) The data has a long-left tail
 D) The data has no skew
13. Which of the following best describes kurtosis in a dataset?
 A) Degree of flatness or peakedness
 B) Central tendency
 C) Measure of spread
 D) Direction of skew
14. In a box plot, the length of the box represents:
 A) The range of the dataset
 B) The standard deviation
 C) The interquartile range (IQR)
 D) The mean of the dataset
15. Generalization error refers to:
 A) The error a model makes on the training data
 B) The difference between predicted and actual values in the training set
 C) The error a model makes on unseen data
 D) The residuals in a regression model
16. If a model has low training error but high generalization error, it indicates:
 A) Overfitting
 B) Cross-validation
 C) Underfitting
 D) Good model selection
17. Which of the following is an example of an out-of-sample evaluation metric?
 A) R-squared on the training set
 B) Confusion matrix on the training data
 C) Residual error on the training set
 D) Mean Squared Error (MSE) on the test set
18. Cross-validation is used to:
 A) Train a model on all the data available
 B) Reduce the number of variables
 C) Reduce the variance in the dataset
 D) Test a model's ability to generalize on different subsets of the data
19. Overfitting occurs when a model:
 A) Fits the noise in the training data instead of the true underlying pattern
 B) Is too simple to capture the patterns in the data
 C) Has a high bias
 D) Has too few parameters
20. Grid search allows for:
 A) Hyperparameter tuning in a systematic way
 B) Improving the accuracy of predictions
 C) Reducing the number of features
 D) Automating the creation of new models

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

1. a) List and explain any 5 common challenges in Data Collection.
 b) Define a **Confusion Matrix** and explain its components. Consider the following confusion matrix for a dataset of size n=100.

n = 100	Actual: YES	Actual: NO
Predicted: YES	60	7
Predicted: NO	8	25

Find:

- | | | | | | |
|------|-------------|---|----|---|---|
| i. | Accuracy | | | | |
| ii. | Error rate | | | | |
| iii. | Precision | | | | |
| iv. | F1- measure | 8 | L2 | 2 | 1 |
2. a) Explain the process of data science in detail. 8 L2 2 1
 b) What is data discretization in data preprocessing, and what are the different techniques used for discretization? 8 L3 2 1
3. a) Explain the 5 V's in big data. 8 L2 1 1
 b) Explain probability sampling and its variants with suitable examples. 8 L2 1 1

Unit – II

4. a) Define Kurtosis and explain its significance in statistics. Differentiate between Platykurtic, Mesokurtic, and Leptokurtic distributions with neat sketches/examples. 8 L2 4 1
- b) Explain the three Types of Exploratory Data Analysis. 8 L2 3 1
5. a) What is the need for polynomial regression and explain the advantages of using EDA. 8 L2 4 1
- b) What is Linear regression. Explain the assumptions underlying Multiple Regression. 8 L2 3 1
6. a) Briefly explain the Key aspects of Exploratory Data Analysis 8 L2 3 1
- b) The data below shows the average daily temperature (in degrees Celsius) for a city during summer over the last 6 days, along with the number of ice creams sold on those days.

Day	Temperature (°C)	Ice Creams Sold
1	25.0	320
2	27.5	450
3	30.0	520
4	32.0	580
5	28.5	490
6	35.0	620

- i. Find the equation of the linear regression line that models the relationship between temperature and the number of ice creams sold.
- ii. Predict how many ice creams will be sold if the temperature reaches 31°C.
- iii. Calculate the coefficient of determination (R^2) for the regression model and interpret its meaning in the context of this data. 8 L3 4 1

Unit – III

7. a) Explain the concept of generalization error . Describe strategies and techniques that can be employed to mitigate generalization error 8 L2 5 1
- b) Explain the concept of cross-validation in machine learning and its importance in model assessment and selection. Compare and contrast three common types of cross-validation techniques 8 L2 5 1
8. a) Suppose you are building a machine learning model for predicting **student exam performance**. Discuss how you would evaluate the model using:
- i. Cross-Validation
- ii. Grid Search for hyperparameter tuning. 8 L2 5 1
- b) Discuss the concepts of overfitting and underfitting in the context of machine learning and data science. Provide examples of when each of these issues may occur and explain the potential consequences of overfitting and underfitting on model performance. 8 L3 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
